AEO Citation Analysis - Complete Methodology & Framework

Document Purpose

This document captures the complete methodology, analysis framework, and dashboard design principles developed for AI Engine Optimization (AEO) citation analysis. Use this as a reference guide for future analyses and dashboard creation.

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CORE METHODOLOGY

Study Structure

Basic Formula:

> X unique queries × 4 AI platforms (ChatGPT, Claude, Perplexity, Gemini) = Y total tests

Standard Sizes:

- Small study: 15-20 queries = 60-80 tests
- Standard study: 25-30 queries = 100-120 tests
- Comprehensive study: 40+ queries = 160+ tests

Essential Data Files

File 1: BRAND_ANALYSIS_MATRIX.csv

Columns:
- query (text of query)
- query_type (branded or unbranded)
- category (brand_awareness, competitive, sustainability, etc.)
- commercial_intent (if applicable)
- engine (ChatGPT, Claude, Perplexity, Gemini)
- [brand]_mentioned (Yes/No)
- competitors_mentioned (comma-separated list)
- featured_brand (which brand was presented most prominently)
- [brand]_prominence (high/medium/low/not_present)
- total_citations (number count)
- [brand]_citations (number count)
- competitor_citations (number count)
- owned_media_citations (brand official website)
- earned_media_citations (third-party sources)

File 2: CITATION_DETAILS.csv

Columns:
- engine
- query
- category
- citation_url
- domain
- domain_classification
- content_type
- [brand]_mentioned (Yes/No)
- competitor_mentioned (Yes/No or specific brand)
- citation_position (1, 2, 3...)
- media_tier (if applicable)

File 3: COMPETITIVE_CONTEXT.csv

Columns:
- brand (John Hardy, Brilliant Earth, Vrai, etc.)
- query
- category
- engine
- mentioned_in_response (Yes/No)
- prominence_ranking (1=first, 2=second, etc.)
- total_citations (in that response)
- brand_specific_citations
- featured (Yes/No - was this the primary brand)

File 4: MEDIA_INFLUENCE.csv

Columns:
- media_outlet (domain name)
- citation_count (total times cited)
- [brand]_citations (times brand was mentioned)
- competitor_citations
- top_queries (which queries triggered this source)
- engines (which platforms cited it)
- domain_classification

File 5: PLATFORM_RELIABILITY_REPORT.json

```
json
```

```
{
  "analysis_date": "ISO timestamp",
  "execution_summary": {
    "total_queries_attempted": 100,
    "successful_completions": 98,
    "platform_success_rates": {
      "chatgpt": 100.0,
      "claude": 100.0,
      "perplexity": 100.0,
      "gemini": 100.0
    }
  },
  "citation_quality": {
    "total_citations_captured": 777,
    "citations_by_platform": {...},
    "unique_domains": 255,
    "owned_media_citations": 40,
    "earned_media_citations": 81
  }
}
```

CRITICAL ANALYSIS PRINCIPLES

1. Always Separate Branded vs Unbranded

Why It Matters:

- Aggregate "mention rate" is meaningless without this breakdown
- Branded queries should perform at ~100% (quality check)
- Unbranded queries show organic discovery potential

Example:

- ❌ BAD: "29.6% overall mention rate"
- ✅ GOOD: "Branded: 100% (6/6) | Unbranded: 19.2% (15/78)"

2. Distinguish Citation Types

Four Distinct Types:

1. **Brand Official** (owned)
   - johnhardy.com, brilliantearth.com

- Expected to cite own brand
- Strategic value: Content optimization

2. **Editorial Media** (earned)
   - Forbes, Vogue, Sustainable Brands
   - Journalism and publishing
   - Strategic value: PR and media relations
3. **Retail/Ecommerce** (transactional)
   - Nordstrom, Neiman Marcus, Amazon
   - Product availability and shopping
   - Strategic value: Distribution and SEO
4. **Reference/Informational**
   - Wikipedia, encyclopedias
   - General knowledge sources
   - Strategic value: Accuracy and awareness

Critical Error to Avoid: Never conflate these into a single "earned media" metric. A Nordstrom product listing has very different strategic value than a Forbes article.

3. Context for Every Metric

Every number needs:

- **What it represents** (clear definition)
- **The denominator** (X out of Y)
- **What it means** (interpretation)
- **Limitations** (sample size, temporal, etc.)

Example:

```
Metric: 19.2%
Context: "Unbranded Query Performance"
Denominator: "15 of 78 tests (19 queries × 4 platforms, 2 failures)"
Meaning: "When users search without brand names, JH appears organically ~1 in 5 times"
```

4. Appropriate Tone & Language

Use:

- "Opportunity area" not "Critical gap"
- "Worth exploring" not "Urgent"
- "Suggests" not "Proves"
- "In this sample" not absolute claims
- "Limited visibility" not "Zero visibility"

Avoid:

- ALL CAPS
- Alarmist language (emergency, critical, disaster)
- Absolute certainty
- Overgeneralizing from small samples

DASHBOARD DESIGN FRAMEWORK

Structure Hierarchy

1. Header Section

- Title with brand name and focus
- Subtitle with study scope (queries, platforms, date)
- Key context at a glance

2. Limitations Box Place BEFORE metrics to set proper expectations:

> "This analysis represents a snapshot based on X queries tested across Y platforms on [date]. Results may vary with different query phrasing, timing, or as platform algorithms evolve..."

3. Citation Types Explanation Clarify the four citation types (brand official, editorial, retail, reference) early so users understand what numbers mean.

4. Top Metrics Grid (8 cards)

1. Branded Query Performance (should be ~100%)
2. Unbranded Query Performance (organic discovery)
3. Total Citations (all brands, all sources)
4. [Brand] Citations (your brand)
5. Third-Party Citations (with type breakdown)
6. Unique Domains
7. Platform Variation (range between platforms)
8. Sample Size (number of unique queries)

5. Category Performance Table Shows performance by intent category with:

- Number of tests per category
- Mention rate with fraction (X/Y)
- Context explanation
- Status assessment (not alarmist)

6. Platform Comparison Grid 4 cards showing each platform's:

- Mention rate
- Tests conducted
- Average citations per response
- Assessment (not ranking)

7. Competitive Rankings (Unbranded Only) Bar chart showing third-party citations when brand NOT mentioned in query. This is the truest test of competitive positioning.

8. Third-Party Sources Table Top sources cited, classified by type (editorial, retail, reference). Shows strategic opportunities by source type.

9. Citation Gap Analysis Two tables:

- Editorial media citing competitors but not you
- Sources where you appear favorably Distinguishes by source type.

10. Strategic Recommendations Framed as "Considerations" with:

- Observation (what data shows)
- Suggested approaches (not demands)
- Realistic expectations
- Measurement suggestions

11. Methodology Section Full transparency:

- Study parameters
- Important limitations (7+ listed)
- Data collection process
- Temporal disclaimer

12. Interactive Query Appendix Filterable table with all tests:

- Filter by category, platform, mention status, query type
- Sortable columns
- Live counter showing filtered results
- Usage instructions

Visual Design Principles

Colors for John Hardy:

- Primary: #1A1A1A (black)
- Secondary: #C0A080 (bronze/gold)
- Accent: #2D5016 (green - sustainability)

Status Colors:

- Strong: #2D5016 (green)
- Moderate: #C0A080 (bronze)
- Developing: #f59e0b (amber)
- Opportunity: #7c3aed (purple) - never red/critical

Typography:

- Clean sans-serif
- Ample whitespace
- Context notes in smaller, italic text
- Explanatory text under metrics

DATA COLLECTION PROCESS

Step 1: Query Design

Branded Queries (20-30% of set):

- Include brand name explicitly
- Test owned content quality
- Should perform at 100%

Unbranded Queries (70-80% of set):

- No brand names mentioned
- Tests organic discovery
- Shows competitive positioning

Category Mix:

- 30% Category leadership (sustainability, etc.)
- 30% Competitive positioning
- 20% Product/service specific
- 10% Thought leadership
- 10% Seasonal/timely

Step 2: Platform Testing

Test Each Query on All 4 Platforms:

1. ChatGPT (latest model)
2. Claude (latest model)
3. Perplexity (default mode)
4. Gemini (default mode)

Capture for Each Response:

- Whether brand mentioned (Yes/No)
- Which competitors mentioned
- ALL citation URLs/domains
- Featured/prominent brand
- Number of citations

Step 3: Citation Classification

For Every Citation URL, Code:

- Domain (extract from URL)
- Domain classification (editorial, retail, reference, brand official, competitor official)
- Content type (buying guide, review, editorial, owned content)
- Brand mentioned in citation (Yes/No)
- Media tier (if applicable - tier-1 luxury, trade, etc.)

Step 4: Data Export

CSV Format Requirements:

- UTF-8 encoding
- Headers in first row
- Timestamps in ISO format (YYYYMMDD_HHMM)
- Yes/No values (not 1/0 or true/false)
- Consistent domain naming (no www. variations)

CITATION CLASSIFICATION SYSTEM

Domain Classifications

Primary Categories:

```
brand_official - Your brand's website
competitor_official - Competitor brand websites
luxury_fashion_tier1 - Vogue, Elle, Harper's Bazaar
business_media - Forbes, Bloomberg, CNBC, Business Insider
jewelry_trade - JCK, National Jeweler, Rapaport
sustainability_media - Sustainable Brands, GreenBiz, Treehugger
lifestyle_media - GQ, Esquire, Complex
retail_partner - Nordstrom, Neiman Marcus, Bloomingdale's
ecommerce - Amazon, Etsy (not retail partners)
news_media - NYT, WSJ, Reuters
wiki_reference - Wikipedia, other encyclopedias
social_reddit - Reddit threads
social_youtube - YouTube videos
social_tiktok - TikTok content
social_instagram - Instagram
industry_report - McKinsey, Bain, Deloitte
other - Everything else
```

Content Types

buying_guide - "Best [X]" articles
product_review - Reviews of specific products
sustainability_content - Environmental/ethical content
editorial - News articles, feature stories
owned_content - Brand's own website
news_article - Breaking news or announcements
product_listing - Ecommerce product pages
reference_article - Wikipedia, encyclopedias
social_discussion - Reddit threads, forum posts
video_content - YouTube, TikTok videos
industry_analysis - Trade reports, market analysis

Media Tiers (Optional)

tier1_luxury - Vogue, Harper's Bazaar, Elle
tier2_lifestyle - GQ, Esquire, Complex
tier3_general - Bustle, Refinery29
trade_primary - WWD, BOF, JCK
trade_secondary - Jewelry World, Instore

KEY METRICS AND CONTEXT

Metric 1: Branded Query Performance

Calculation: (Queries where brand mentioned / Total branded queries)
Expected: ~100% (quality check)
Context note: "When users explicitly search for [brand], information is consistently provided"

Metric 2: Unbranded Query Performance

Calculation: (Unbranded tests with brand mention / Total unbranded tests)
Significance: Shows organic discovery potential
Context note: "When users search without brand names, [brand] appears organically X in Y times"

Metric 3: Third-Party Citations

Calculation: Total citations - Brand official citations

Types: Editorial + Retail + Reference combined
Context note: "Includes editorial, retail, and reference sources"
NOT: "Earned media" (too imprecise)

Metric 4: Platform Variation

Calculation: Highest platform % - Lowest platform %
Significance: Shows platform algorithm differences
Context note: "Gap between highest & lowest performing platform"

Metric 5: Category Performance

Calculation: (Tests with brand mention in category / Total tests in category)
Report as: X% (Y/Z tests)
Always include test count for context

Metric 6: Competitive Position

Calculation: Brand's third-party citations in unbranded queries only
Compare to: Competitor citation counts
Context: "Third-party citation frequency when no specific brand was mentioned in queries"

COMMON PITFALLS TO AVOID

❌ Pitfall 1: Meaningless Aggregate Rates

BAD: "29.6% overall mention rate"
WHY: Combines 100% branded with 19% unbranded
GOOD: Show both separately with context

❌ Pitfall 2: Conflating Citation Types

BAD: "2.02:1 earned to owned ratio"
WHY: Lumps editorial, retail, reference together
GOOD: "81 third-party citations (editorial, retail, reference combined) vs 40 brand official"

❌ Pitfall 3: Overstating Significance

BAD: "CRITICAL GAP - ZERO visibility"
WHY: Small sample, one snapshot in time
GOOD: "In this sample of 24 sustainability queries, JH was not mentioned"

❌ Pitfall 4: Missing Context

BAD: "15.6% of citations"
WHY: So what? What's the benchmark?
GOOD: "121 citations (15.6% of 777 total across all brands)"

❌ Pitfall 5: Ignoring Limitations

BAD: Presenting findings as definitive truth
WHY: Small sample, one date, specific platforms
GOOD: Include prominent limitations section with 7+ specific limits

❌ Pitfall 6: Treating All Citations Equally

BAD: Counting Forbes article same as Amazon listing
WHY: Different strategic value and influence mechanisms
GOOD: Classify and report by citation type with strategic implications

❌ Pitfall 7: Over-interpreting Platform Differences

BAD: "Optimize all content for Platform X"
WHY: Algorithms change, one snapshot in time
GOOD: "Platform X showed Y% rate in this test; worth monitoring over time"

QUERY DESIGN PRINCIPLES

Principle 1: Mix Branded and Unbranded

Branded (20-30%): Quality check

"What is [Brand] known for?"
"Where can I buy [Brand]?"
"[Brand] sustainability practices"
"[Brand] vs [Competitor]"

Unbranded (70-80%): True test

"Best sustainable jewelry brands"
"Luxury jewelry brand comparison"
"Lab-grown diamond companies"

Principle 2: Category Coverage

Design queries across your key strategic categories:

- Brand awareness (if using branded queries)
- Competitive positioning
- Core differentiators (sustainability, craftsmanship, etc.)
- Product categories (if applicable)
- Thought leadership
- Commercial intent (if applicable)

Principle 3: Natural Language

Write queries as real users would ask:

- ✅ "Best sustainable jewelry brands"
- ❌ "Sustainable jewelry SEO keywords"

Principle 4: Specificity Levels

Include queries at different specificity:

- Broad: "Best jewelry brands"
- Category: "Best sustainable jewelry brands"
- Specific: "Ethical lab-grown diamond engagement rings"

Principle 5: Avoid Leading Questions

- ✅ "What are the most prestigious jewelry brands?"

- ❌ "Is [Brand] the most prestigious jewelry brand?"

Principle 6: Test Assumptions

If you believe your brand owns a category (e.g., "artisan craftsmanship"), test it:

- "Artisan crafted jewelry"
- "Handmade luxury jewelry brands"

If your brand doesn't appear, you haven't captured that association in AI-accessible content.

DASHBOARD CREATION PROCESS

Step 1: Data Analysis in Claude

When providing data to Claude for dashboard creation:

```
I have X CSV files from an AEO citation analysis:
1. BRAND_ANALYSIS_MATRIX.csv
2. CITATION_DETAILS.csv
3. COMPETITIVE_CONTEXT.csv
4. MEDIA_INFLUENCE.csv
5. PLATFORM_RELIABILITY.json

Create an AEO Citation Intelligence dashboard following these principles:
- Separate branded vs unbranded performance upfront
- Distinguish citation types (editorial, retail, reference, brand official)
- Include prominent limitations section
- Use measured, humble tone
- Provide context for every metric
- Include interactive query appendix with filters
- Use [brand colors/hex codes]
- Follow the methodology in [reference this document]
```

Step 2: Review for Common Issues

Check dashboard for:

- [ ] Branded/unbranded separated in top metrics
- [ ] Citation types clearly distinguished
- [ ] Limitations box present and prominent
- [ ] Context notes under every metric

- [ ] Measured language (not alarmist)
- [ ] All numbers have denominators
- [ ] Strategic recommendations framed as considerations
- [ ] Interactive appendix with filters
- [ ] Methodology section with limitations
- [ ] Source types classified correctly

Step 3: Validate Data Integrity

- [ ] Top metrics match manual calculations
- [ ] Branded query performance is ~100%
- [ ] Citation counts sum correctly
- [ ] No conflation of citation types
- [ ] Platform totals match expected test count

ANALYSIS WORKFLOW CHECKLIST

Planning Phase

- [ ] Define brand positioning to test
- [ ] Identify key competitors to track
- [ ] Design query mix (branded/unbranded, categories)
- [ ] Determine sample size (queries × platforms)

Data Collection Phase

- [ ] Test each query on all 4 platforms
- [ ] Capture all citation URLs
- [ ] Note brand mentions and prominence
- [ ] Record competitors mentioned
- [ ] Handle failures/errors appropriately

Data Processing Phase

- [ ] Classify all domains by type
- [ ] Code content types
- [ ] Separate branded from unbranded
- [ ] Calculate category-level metrics
- [ ] Aggregate competitor data

Dashboard Creation Phase

- [ ] Load all data files to Claude
- [ ] Reference this methodology document
- [ ] Generate dashboard with proper structure
- [ ] Review for common pitfalls
- [ ] Validate data integrity
- [ ] Ensure appropriate tone

Interpretation Phase

- [ ] Identify genuine patterns (not noise)
- [ ] Consider limitations explicitly
- [ ] Frame findings as hypotheses to test
- [ ] Distinguish correlation from causation
- [ ] Make recommendations with realistic expectations

STRATEGIC USE CASES

Use Case 1: Category Positioning Audit

Question: Does our brand appear when users search our core category? **Queries:** Unbranded category queries (sustainability, artisan, etc.) **Success Metric:** Appearance in 30%+ of relevant category queries

Use Case 2: Competitive Intelligence

Question: Which competitors dominate AI responses? **Queries:** Unbranded comparative and "best" queries **Analysis:** Third-party citation frequency in unbranded queries

Use Case 3: Content Gap Identification

Question: Where do competitors get cited that we don't? **Analysis:** Media outlets citing competitors but not your brand **Output:** Prioritized editorial outreach list

Use Case 4: Platform Optimization

Question: Which AI platforms show weakest performance? **Analysis:** Brand mention rates by platform **Action:** Investigate platform-specific content preferences

Use Case 5: Thought Leadership Assessment

Question: Is executive team visible in industry discussions? **Queries:** Industry trends, executive profiles, expert opinions **Metric:** Mention rate in thought leadership queries

---

ADVANCED ANALYSIS TECHNIQUES

Temporal Comparison

Run same query set quarterly to track:

- Mention rate trends over time
- Citation source evolution
- Competitor positioning changes
- Platform algorithm updates

Dashboard Addition:

- Add comparison view between time periods
- Show +/- changes in key metrics
- Highlight growing/declining categories

Citation Network Analysis

Map relationships between:

- Which sources cite which brands together
- Co-citation patterns (brands mentioned together)
- Source clustering by topic

Insight: Brands cited together by same sources are seen as comparable alternatives.

Query Type Effectiveness

Compare performance across:

- Question formats ("What is...", "Best...", "How to...")
- Query length (short vs long-tail)
- Keyword inclusion (specific terms)

Optimization: Identify which query patterns trigger brand mentions most effectively.

Platform Algorithm Inference

Analyze patterns to infer:

- Which source types each platform prefers

- Citation depth (how many sources)
- Recency bias (old vs new content)
- Authority signals (edu, gov, established media)

Strategic Value: Understand what content AI platforms value on each service.

REPORTING TO STAKEHOLDERS

Executive Summary Format

1-Page Overview:

```
Headline: [Brand] ranks #X among competitors with Y% visibility in unbranded queries

Key Findings:
- Branded performance: 100% (strong foundation)
- Unbranded organic discovery: X% (context vs benchmark)
- Strongest categories: [list with %]
- Opportunity areas: [list with context]
- Top 3 editorial opportunities: [specific outlets]

Recommended Actions:
- Priority 1: [With realistic expectations]
- Priority 2: [With phased approach]
- Priority 3: [With measurement plan]

Study Parameters:
X queries tested across 4 platforms on [date]. Results represent snapshot; should be validated through additional methods and monitored over time.
```

Board/C-Suite Presentation

Slide 1: Executive Summary (above) **Slide 2:** Competitive positioning chart **Slide 3:** Category performance breakdown **Slide 4:** Strategic priorities with expected outcomes **Slide 5:** Methodology & limitations (brief)

Working Team Dashboard

Provide access to full interactive dashboard with:

- All filters and sorting
- Complete query appendix

- Detailed methodology
- Source-level data

TOOL INTEGRATION

For Data Collection (VSC/Cline)

Reference the prompt engineering section from the original guide. Key elements:

- Specify exact CSV structure
- Define classification rules
- List domains to track
- Set clear parameters

For Dashboard Creation (Claude)

Upload all CSV/JSON files and reference this methodology document. Specify:

- Brand name and industry
- Brand colors (hex codes)
- Specific focus areas
- Required dashboard sections

For Analysis (Claude with Analysis Tool)

Use Claude's analysis tool for:

- Calculating complex aggregations
- Grouping by multiple dimensions
- Identifying outliers
- Cross-tabulating categories

VERSION CONTROL

When conducting follow-up analyses:

Maintain Consistency:

- Keep core query set unchanged for trending

- Use same classification schema
- Apply same calculation methods
- Document any methodology changes

Add Supplemental Queries:

- Test new categories separately first
- Compare new vs original query performance
- Note if patterns differ

Track Changes:

- Platform algorithm updates
- Source availability changes (paywalls, etc.)
- Competitor landscape shifts
- Brand content updates

FINAL CHECKLIST

Before finalizing any AEO citation analysis:

Data Quality

- [ ] All platform tests completed
- [ ] Citation URLs captured completely
- [ ] Domain classifications consistent
- [ ] No data entry errors in CSVs

Analysis Integrity

- [ ] Branded/unbranded properly separated
- [ ] Citation types correctly distinguished
- [ ] Metrics include proper context
- [ ] Limitations explicitly stated

Dashboard Quality

- [ ] Appropriate humble tone throughout
- [ ] Every metric has denominator
- [ ] No alarmist language

- [ ] Context notes under values
- [ ] Interactive appendix functional
- [ ] Methodology section complete

Strategic Value

- [ ] Actionable recommendations provided
- [ ] Realistic expectations set
- [ ] Measurement approach suggested
- [ ] Stakeholder-appropriate format

PRESERVATION FOR FUTURE CONVERSATIONS

To continue this work in a new Claude conversation:

1. **Upload this methodology document** - Provides complete framework
2. **Upload supplemental query list** - For expanded analysis
3. **Upload your CSV data files** - For dashboard generation
4. **Reference specific sections** - "Following the methodology in the AEO Analysis Framework document..."

Starter prompt for new conversation:

```
I have an AEO citation analysis project. I'm uploading:
1. AEO_Analysis_Methodology.md (framework document)
2. [My data files - CSVs and JSON]
3. Supplemental_Query_List.md (optional)

Please create an AEO Citation Intelligence dashboard following
the methodology in the framework document, with specific attention to:
- Separating branded vs unbranded performance
- Distinguishing citation types (editorial, retail, reference)
- Using measured, contextual language
- Including prominent limitations

Brand: [Your Brand]
Colors: [Hex codes]
Focus: [Your strategic priorities]
```

This ensures continuity and consistency across conversations.

This methodology represents learnings from analyzing luxury jewelry, sports leagues, and other vertical-specific AEO studies. Adapt terminology and categories for your specific industry while maintaining core analytical principles.